

Deliverable 2: Bayesian Spatial Search

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1 Prior distribution

1.1 Mean: two-phase displacement model

The object falls from the aircraft at explosion point $\mathbf{x}_E = (7, 20)$. Initially, it only carries the aircraft's velocity $\mathbf{v}_{\text{plane}}$. After this, it experiences two phases before reaching the seafloor.

Aerial Phase The object is subject to the aerodynamic drag from the air, that moves at $\mathbf{v}_{\text{wind}}$. The drag force $\mathbf{F}_{\text{drag}} = -(m/\tau)(\mathbf{v}(t) - \mathbf{v}_{\text{wind}})$ gives the ODE $\dot{\mathbf{v}} = -(1/\tau)(\mathbf{v}(t) - \mathbf{v}_{\text{wind}})$, whose solution is

$$\mathbf{v}(t) = \mathbf{v}_{\text{wind}} + (\mathbf{v}_{\text{plane}} - \mathbf{v}_{\text{wind}})e^{-t/\tau}.$$

Supposing $T_{\text{air}} \ll \tau$ (heavy object, short fall) and approximating $e^{-t/\tau} \approx 1 - t/\tau$ (Taylor expansion), integrating over $[0, T_{\text{air}}]$ gives

$$\Delta \mathbf{x}_{\text{air}} \approx \mathbf{v}_{\text{plane}} T_{\text{air}} + \frac{1}{2}(\mathbf{v}_{\text{wind}} - \mathbf{v}_{\text{plane}}) \frac{T_{\text{air}}^2}{\tau}.$$

The $\frac{1}{2}$ factor comes from the object being fully decoupled from the wind at $t = 0$ and gradually acquiring wind velocity through drag. The average effective coupling over the aerial phase is half the final value.

Aquatic phase Water drag is $\approx 800\times$ stronger than air drag, so the object couples quickly to $\mathbf{v}_{\text{drift}}$ after it impacts the water. The water drift velocity applies for most of T_{drift} , giving $\Delta \mathbf{x}_{\text{drift}} \approx \mathbf{v}_{\text{drift}} T_{\text{drift}}$ with no $\frac{1}{2}$ factor.

Weight derivation. Collecting terms and normalizing by T_{air} (setting $w_{\text{plane}} = 1$):

$$w_{\text{plane}} = 1, \quad w_{\text{wind}} = \frac{T_{\text{air}}}{2\tau} < \frac{1}{2}, \quad w_{\text{drift}} = \frac{T_{\text{drift}}}{T_{\text{air}}} < w_{\text{wind}}.$$

We set $w_{\text{wind}} = 0.4$ and $w_{\text{drift}} = 0.3$, which satisfies both inequalities. Witness 1 confirmed a plane-dominated motion, so the plane contribution is set to be $5.81\times$ larger than wind + drift combined ($\|\mathbf{v}_{\text{plane}}\| = 6.95$ vs. $0.4 \times 1.80 + 0.3 \times 1.58 = 1.19$). We can check the robustness of the selected weights modifying each one ± 0.1 . We observe that it shifts $\boldsymbol{\mu}$ by less than 0.4 grid units.

The prior mean is:

$$\boldsymbol{\mu} = \mathbf{x}_E + w_{\text{plane}} \mathbf{v}_{\text{plane}} + w_{\text{wind}} \mathbf{v}_{\text{wind}} + w_{\text{drift}} \mathbf{v}_{\text{drift}} = (12.75, 15.45).$$

1.2 Covariance: anisotropic Gaussian

The Gaussian is the maximum-entropy distribution for a given mean and covariance. This makes it the least informative choice consistent with the first two moments we can estimate. The covariance is anisotropic because the main source of uncertainty is along-track: unknown altitude h determines $T_{\text{air}} = \sqrt{2h/g}$, and unknown drag timescale τ affects how far the object travels along the flight direction. Lateral deviations are smaller, constrained by Witness 2 (*“slightly off to one side”*). We use $\sigma_{\parallel} = 8$ and $\sigma_{\perp} = 4$ (grid units), with axes aligned to the aircraft direction: $\Sigma = \mathbf{R} \text{diag}(\sigma_{\parallel}^2, \sigma_{\perp}^2) \mathbf{R}^{\top}$, where $\mathbf{R} = [\hat{\mathbf{v}}_{\text{plane}} \mid \hat{\mathbf{v}}_{\text{plane}}^{\perp}]$. The 1σ ellipse covers 13% of grid cells.

2 Detection model

We use the Koopman negative-exponential model:

$$\rho_{tj} = 1 - \exp(-e_t \kappa_j), \quad \kappa_j = \kappa_0 (1 - \alpha_d d_j) (1 - \alpha_r r_j),$$

where $d_j, r_j \in [0, 1]$ are normalized depth and roughness.

Functional form The negative-exponential is the *unique* form satisfying: (i) $\rho \rightarrow 0$ as $e \rightarrow 0$; (ii) $\rho \rightarrow 1$ as $e \rightarrow \infty$; (iii) strict monotonicity; (iv) diminishing returns ($\partial^2 \rho / \partial e^2 < 0$). It comes from the Poisson detection process: if detections follow a Poisson process with rate κ_j per unit effort, then $P(\text{miss after effort } e_t) = e^{-\kappa_j e_t}$. The multiplicative form for κ_j is the unique form that stays positive for all $(d_j, r_j) \in [0, 1]^2$ and captures the interaction between depth and roughness.

Parameter calibration. Rather than choosing parameters arbitrarily, we derive four inequality constraints from the previous mission reports and select the *maximum-margin solution*. This is the parameter triple that maximizes the minimum slack to any constraint boundary.

Mission	Conditions	Report outcome	Constraint
M1	$e = 1$, moderate d , r	No detect, “not strong evidence”	$\rho < 0.50$
M2	$e = 3$, shallow, smooth	Detected quickly	$\rho > 0.60$
M3	$e = 3$, deep, rough	No detect, “not surprising”	$\rho < 0.25$
M4	$e = 2$, moderate	Detected after repeated inspection	$\rho > 0.30$

Grid search yields $\kappa_0 = 0.65$, $\alpha_d = 0.90$, $\alpha_r = 0.75$ (margin = 0.083). Depth is penalized more than roughness ($\alpha_d > \alpha_r$) because Mission 3 identifies “*strong depth effects*” as the primary cause of failure.

3 Posterior update and mission results

The posterior is computed sequentially in log-space:

$$\log \pi_j^{(T)} = \log \pi_j + \sum_{t=1}^T \log P(s_t \mid Z = j) + \text{const}, \quad P(s_t = 0 \mid Z = j) = 1 - c_{tj} \rho_{tj}.$$

Log-space avoids underflow: the product of likelihood terms can reach 10^{-50} or smaller with only a few missions. The log-sum-exp shift is applied before exponentiation to guarantee numerical stability. Missions are assumed conditionally independent given Z .

We submitted three missions to the Streamlit app following the budget constrains. They all returned $s_t = 0$ (no detection).

Mission	Region $[x, y]$	e	n	Cost	s_t	Prior mass	Post. mass	Reduction
M1	$[7, 19] \times [12, 18]$	1	72	72	0	0.298	0.187	37.4%
M2	$[10, 16] \times [9, 21]$	1	72	72	0	0.275	0.193	29.9%
M3	$[6, 20] \times [12, 18]$	1	84	84	0	0.332	0.218	34.3%

Total cost: 228/230. Each no-detection reduces posterior mass inside the searched region by a factor proportional to $(1 - q_{tj})$. Cells with low κ_j (deep, rough) receive smaller reductions.

4 Proposed next mission

We try to maximize the Expected Probability of Detection (EPD). It has been chosen over entropy reduction because the goal is *detection*, not localization:

$$\text{EPD}(R, e) = \sum_{j \in R} \pi_j^{(T)} \rho(e, d_j, r_j), \quad \text{s.t.} \quad e \cdot |R| \leq B_{\text{rem}}.$$

The proposed mission is the EPD-optimal region computed from $\pi_j^{(3)}$ assuming an additional budget of 50 units were available. The three effort-1 missions we selected achieve a combined EPD of $1 - (1 - 0.069)(1 - 0.059)(1 - 0.061) = 17.8\%$, which exceeds the 17.2% of a single effort-2 mission; the cross-over only favors higher effort when the posterior is concentrated in fewer than ~ 20 cells.

We observe that the three regions overlap substantially: M1 is completely contained within M3 (100% overlap), and M1 and M2 share 50% of their cells. The union covers only 120 cells (6.9% of the grid). This occurred because each mission was centered on the posterior mean, which remained near the same high-prior region.

Maybe a coverage-constrained design (penalize overlap) would achieve more spatial coverage for the same budget.

5 Discussion

Strengths. (1) Weights derived analytically from the drag ODE; the $\frac{1}{2}$ is a direct consequence of linear coupling, not a heuristic. (2) Detection parameters calibrated via maximum-margin constraints. (3) Both functional forms (negative-exponential, multiplicative κ_j) are the unique choices satisfying their respective physical requirements. (4) Log-space implementation is numerically stable. (5) EPD directly optimises the search objectiv

Limitations. (1) T_{air} and τ are unknown. The weights cannot be precisely calibrated without altitude data; $w_{\text{wind}} = 0.4$ is consistent with the bound $< \frac{1}{2}$ but not uniquely determined. (2) Mission report constraints are qualitative, not measurements; the four inequalities are soft. (3) Gaussian prior has infinite support. (4) Conditional independence may be violated by shared equipment failures. (5) Search was inconclusive: the budget was insufficient to cover the full posterior support with high detection probability.